Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2019)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Which discretization method is used to obtain the discrete-time system?	**Forward Euler (explicit Euler) discretization**	*The difference equation y[n] = (1 + a Ts) y[n-1] + Ts u[n-1] comes from approximating dy/dt by (y[n] - y[n-1])/Ts and evaluating the right-hand side at the previous sample, which is the forward Euler method.*
q2	Determine the z-transform variable as a function of s, i.e., a relation of the form z = f(s).	**z = 1 + s Ts**	*Z-transforming the difference equation gives H(z) = Ts/(z - 1 - a Ts). Matching this to the sampled continuous transfer function H(s) = 1/(s - a) via s = (z - 1)/Ts yields H(z) = Ts/(z - 1 - a Ts). Solving for z gives z = 1 + s Ts, the forward Euler mapping.*
q3	Mapping of continuous-time stable poles to z-plane via forward Euler discretization	**The stable continuous-time poles (Re(s)<0) map to the region in the z-plane where Re(z)<1, i.e., the half-plane left of the vertical line through z=1.**	*Using the forward Euler relation s ≈ (z-1)/Ts, the condition Re(s)<0 becomes Re((z-1)/Ts)<0 => Re(z-1)<0 => Re(z)<1. Thus the left half-plane maps to the half-plane left of Re(z)=1.*
q4	Output y2[n] of the given system for input u[n]=1+cos(π n)	**Plot D**	*Compute u[n]=2 for even n, 0 for odd n. The filter y1[n]=0.5 y1[n-1]+0.5 u[n] yields y1[n]=1+(1/3)(-1)^n after transients. Then y2[n]=u[n]-y1[n]=(2/3)(-1)^n, an alternating sequence converging to ±2/3. Starting from initial rest gives y2[-1]=0, then +1, -0.5, +0.75, -0.625, … matching Plot D.*
q5	Find a and b in H(z)=b/(1+az^{-1}) given closed-loop DC gain 1/2 and gain at ω=π 3/2 with K=3.	**a = 2, b = 1**	*Closed-loop T(z)=Kb/(1+az^{-1}+Kb). At z=1: Kb/(1+a+Kb)=1/2. At z=-1: Kb/(1-a+Kb)=3/2. With K=3, solving yields Kb=3 => b=1 and a=2.*
q6	Given b=2, find a for which closed-loop system with K=3 is BIBO stable.	**-7 < a < 7**	*Closed-loop pole at z=-a/7. For causal BIBO stability need |pole|<1 => |a|/7<1 => |a|<7.*
q7	Design causal stable LTI filter G to pass DC unchanged, eliminate cos(pi/2 n) term, and double the (-1)^n term.	**G(z) = (2/3)(1 + z^{-2}) / (1 + (1/3) z^{-1}) or equivalently y[n] = (2/3)(x[n] + x[n-2]) - (1/3) y[n-1]**	*Place zeros at z = ±j to null the π/2 frequency, choose a real pole at p = -1/3 inside unit circle to adjust gains: solve H(1)=1, H(-1)=2 yields A=2/3, p=-1/3. This filter is causal and stable.*
q8	Find LTI system G such that input u[n]=n s[n-1] yields output y[n]=s[n-2].	**u feeds two parallel blocks, -z^(-2) and z^(-1), whose outputs are summed to give y**	*Compute Z-transforms: U(z)=z/(z-1)^2, Y(z)=1/(z(z-1)). Then G(z)=Y/U=z^{-1}-z^{-2}=z^{-1} - z^{-2}, corresponding to a delay of one sample minus a delay of two samples.*
q9	Magnitude and phase response of H(z)=0.25(z+2+z^{-1}) for Ω in [0,π]	**Magnitude: |H(e^{jΩ})| = 0.5(1+cosΩ); Phase: ∠H(e^{jΩ}) = 0 (for Ω where magnitude>0, undefined at Ω=π)**	*Substitute z=e^{jΩ}: H(e^{jΩ})=0.25(e^{jΩ}+2+e^{-jΩ})=0.25(2cosΩ+2)=0.5(1+cosΩ). This is real and nonnegative, so magnitude equals the expression and phase is zero (except at Ω=π where magnitude zero).*
q10	Determine if the system is FIR or IIR, causal or non-causal	**FIR and non-causal**	*H(z)=0.25z+0.5+0.25z^{-1} has a finite number of terms → FIR. Inverse Z-transform gives h[n]=0.25δ[n+1]+0.5δ[n]+0.25δ[n-1], which includes a nonzero sample at n=-1 → depends on future input → non-causal.*
q11	Given H(z)=0.25(z+2+z^{-1}) and input u[n]=cos(pi/3 n), find output y[n].	**y[n] = 0.75 cos(pi/3 n)**	*Frequency response H(e^{jω}) = 0.25(e^{jω}+2+e^{-jω}) = 0.5(1+cos ω). At ω=π/3, cos ω=0.5 => H=0.5(1+0.5)=0.75. Since input is a cosine, output is same frequency scaled by magnitude and zero phase: y[n]=0.75 cos(π/3 n).*
q12	For causal LTI system G, input u[n]=s[n]-s[n-2] yields output y[n]=δ[n-1]+δ[n-4]. Find impulse response h[n].	**h[n] = δ[n-1] + δ[n-3]**	*Input u[n] is a length■2 rectangular pulse: u[n]=1 for n=0,1 else 0, with Z■transform U(z)=1–z^{–2}. Output Y(z)=z^{–1}+z^{–4}. System function H(z)=Y(z)/U(z)=(z^{–1}+z^{–4})/(1–z^{–2})=z^{–1}(1+z^{–3})/(1–z^{–2}). Expanding as a power series gives h[n]=δ[n–1]+δ[n–3], which is causal and satisfies the given input■output relation.*
q13	Determine BIBO stability of causal LTI system given input u[n]=s[n]-s[n-2] and output y[n]=δ[n-1]+δ[n-4].	**Stable**	*From y[n]=h[n]+h[n-1] we solve for h[n] with causality: h[0]=0, h[1]=1, h[2]=-1, h[3]=1, h[n]=0 otherwise. Sum |h[n]|=3 < ∞, so system is BIBO stable.*
q14	If input is a complex exponential sequence, does output have the same frequency?	**True**	*Complex exponentials are eigenfunctions of LTI systems: output = H(Ω)·input, preserving frequency Ω.*
q15	Sum of two cosines input to LTI system; does output equal single cosine at sum frequency?	**False**	*LTI systems cannot generate new frequencies; each cosine scales and shifts phase at its own frequency, so output is sum of two cosines at Ω1 and Ω2, not at Ω1+Ω2.*
q16	Complex exponential input to cascade of two LTI systems; is output given by product of magnitudes and sum of phases?	**True**	*For LTI systems, eigenfunction e^{jΩn} yields output H(Ω)e^{jΩn}; cascade multiplies frequency responses: H=H1H2, giving magnitude product and phase sum.*
q17	True/false statement about LTI system output for input (-1)^n + cos(Ω n)	**False**	*The (-1)^n term corresponds to frequency π, producing output |H(π)|cos(π n+Θ_H(π)), not a DC term |H(0)|cos(Θ_H(0)). Hence the statement is false.*

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q18	DFS coefficients of sampled cosine signals x_i(t)=cos(w_i t+π/4) with periods T_i and sampling Ts_i	**For i=1 (T1=3, Ts1=1, N1=3): X1[1]=½ e^{jπ/4}, X1[2]=½ e^{-jπ/4}, others zero. For i=2 (T2=16, Ts2=5, N2=4): samples [√2/2, -cos(π/8), 0, cos(π/8)] giving X2[0]=√2/8, X2[1]=√2/8 + j·(cos π/8)/2, X2[2]=√2/8, X2[3]=√2/8 - j·(cos π/8)/2. For i=3 (T3=16, Ts3=9, N3=2): samples [√2/2, -cos(3π/8)] giving X3[0]=(√2 - √(2-√2))/4, X3[1]=(√2 + √(2-√2))/4.**	*The continuous signal has w_i=2π/T_i. Sampling yields x_i[n]=cos(2π n Ts_i/T_i + π/4). The DFS coefficients are computed via the DFT of one period of samples.*
q19	Determine whether aliasing occurs for each sampled cosine signal given its period and sampling interval.	**Aliasing occurs only for i = 3.**	*The signal frequency is f_i = 1/T_i Hz. Sampling frequency is Fs_i = 1/Ts_i Hz. Aliasing occurs if Fs_i < 2 f_i, i.e., Ts_i > T_i/2. For i=1: T1=3, Ts1=1 ≤ 1.5 → no aliasing. i=2: T2=16, Ts2=5 ≤ 8 → no aliasing. i=3: T3=16, Ts3=9 > 8 → aliasing.*
q20	Find the DFS coefficients of the output y[n] of the LTI system y[n] = 0.5 y[n-2] + u[n] - u[n-2] with input u[n] = 1 + 0.5 sin(π/2 n) + 0.25 cos(π n).	**Y[0]=0, Y[1]= -j/3, Y[2]=0, Y[3]= j/3.**	*The system frequency response is H(e^{jω}) = (1 - e^{-j2ω})/(1 - 0.5 e^{-j2ω}). The input is periodic with N=4; its DFS coefficients are U[0]=1, U[1]= -j/4, U[2]=1/4, U[3]= j/4. Multiplying: H(0)=0 → Y[0]=0; H(π/2)=4/3 → Y[1]=(4/3)(-j/4)= -j/3; H(π)=0 → Y[2]=0; H(3π/2)=4/3 → Y[3]=(4/3)(j/4)= j/3.*
q21	Magnitude response of discrete-time system y[n] = 0.5 y[n-2] + u[n] - u[n-2]	**Plot D**	*System transfer function H(z)=1/(1-0.5 z^{-2}) has poles at z=±√0.5 and zeros at origin, giving magnitude peaks at Ω=0 and π and a notch at Ω=π/2, which matches a bandstop (notch) filter plot.*
q22	Approximation of system G using lowpass filter W(z) with given magnitude response	**H(z) ~ W(z) W(-z)**	*H(z)=1/(1-0.5 z^{-2}) can be factored as A(z)A(-z) with A(z)=1/(1-√0.5 z^{-1}), a lowpass filter. Hence H(z) is approximated by the product of W(z) and its frequency■shifted version W(-z).*
q23	Find the matrix entries Φ_{i,l} such that y = Φ b implements the inverse DFT.	**[Φ]_{i,l} = (1/N) * exp(j * 2π * i * l / N) for i,l = 0,…,N–1**	*The inverse DFT gives x[n] = (1/N) Σ_{k=0}^{N-1} X[k] e^{j2π kn/N}. Matching y_i = x[i] and b_l = X[l] yields Φ_{i,l} = (1/N) e^{j2π i l / N}.*
q24	Show that the mapping y = Φ b is linear.	**For any scalars α, β and vectors b1, b2, Φ(α b1 + β b2) = α Φ b1 + β Φ b2, so y = α y1 + β y2.**	*Matrix multiplication satisfies Φ(α b1 + β b2) = α Φ b1 + β Φ b2 because Φ is a constant matrix. Hence the transformation is linear, preserving addition and scalar multiplication.*